Logic Basics

Original slides by: Jarrett Billingsley
Modified with bits from: Bruce Childers, David Wilkinson

CS 0447
Introduction to
Computer Programming

Luís Oliveira

Fall 2020

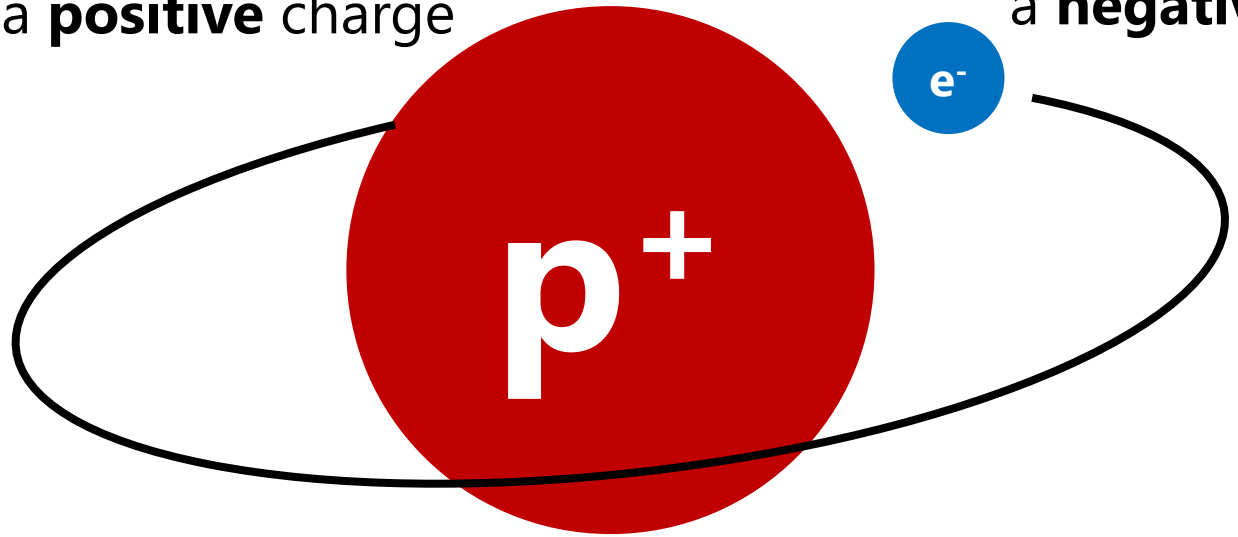
What's electricity?

(EE-ish section)

In your orbit

here's a proton. it has a **positive** charge

here's an electron. it has a **negative** charge

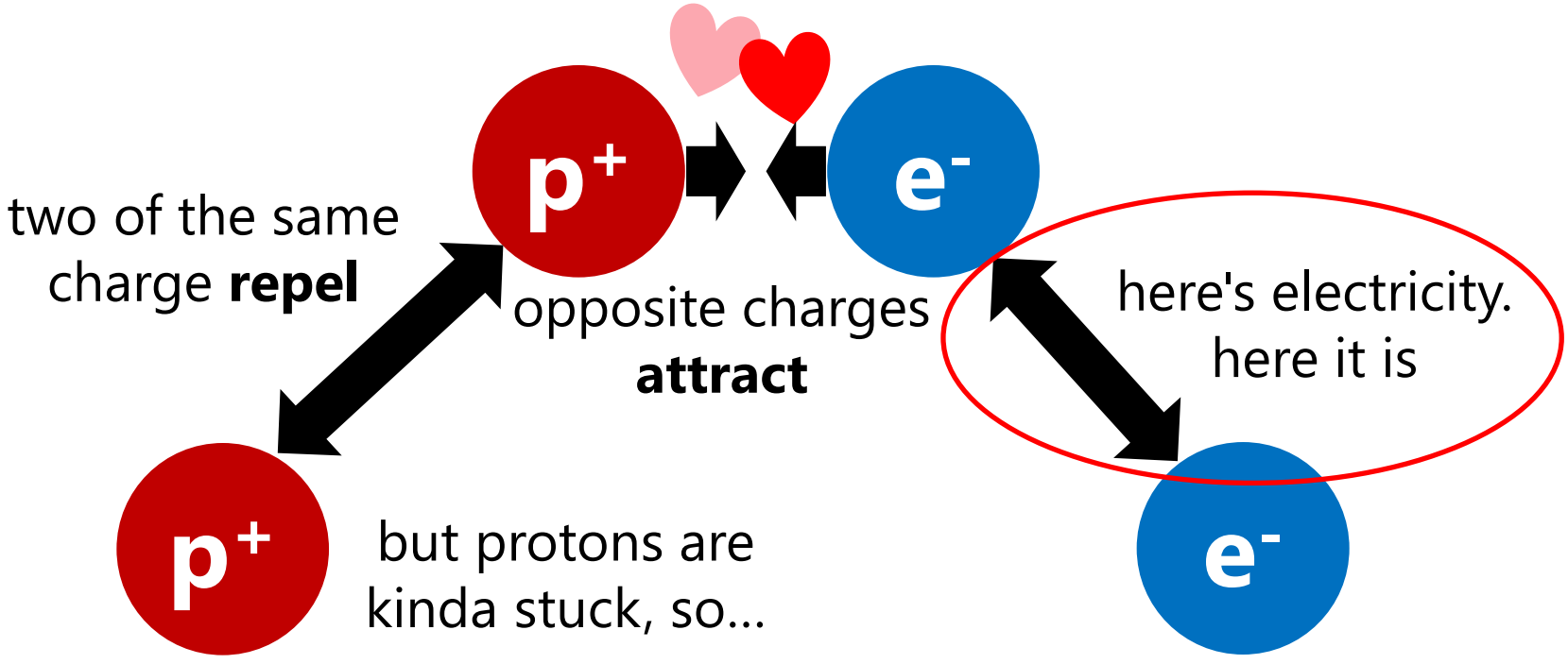


it kinda goes around the proton in an orbit*

protons sit still while electrons can move around

**it actually doesn't... it's way more complex than that! Waves, teleportation and other black magic!*

Opposites attract...

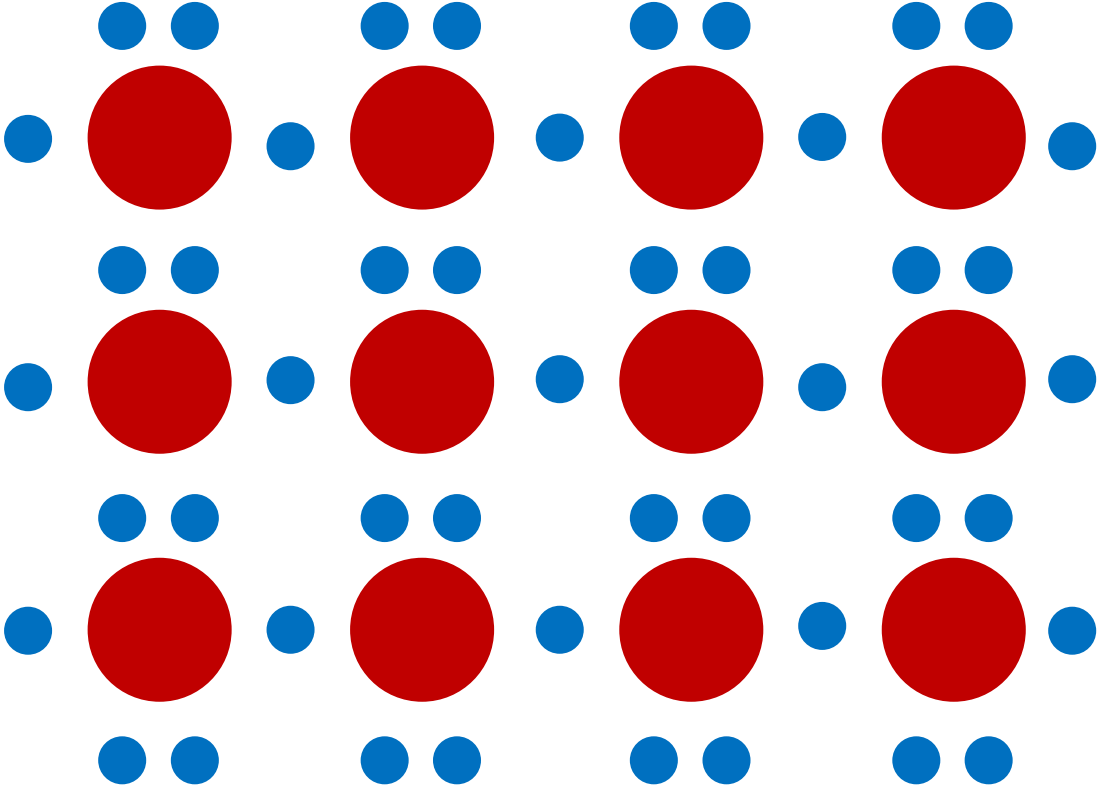


Lots of ~~fish~~ electrons in the sea

here's a solid piece of metal

the atoms are in a fixed structure

but *some* of the electrons are free to move around

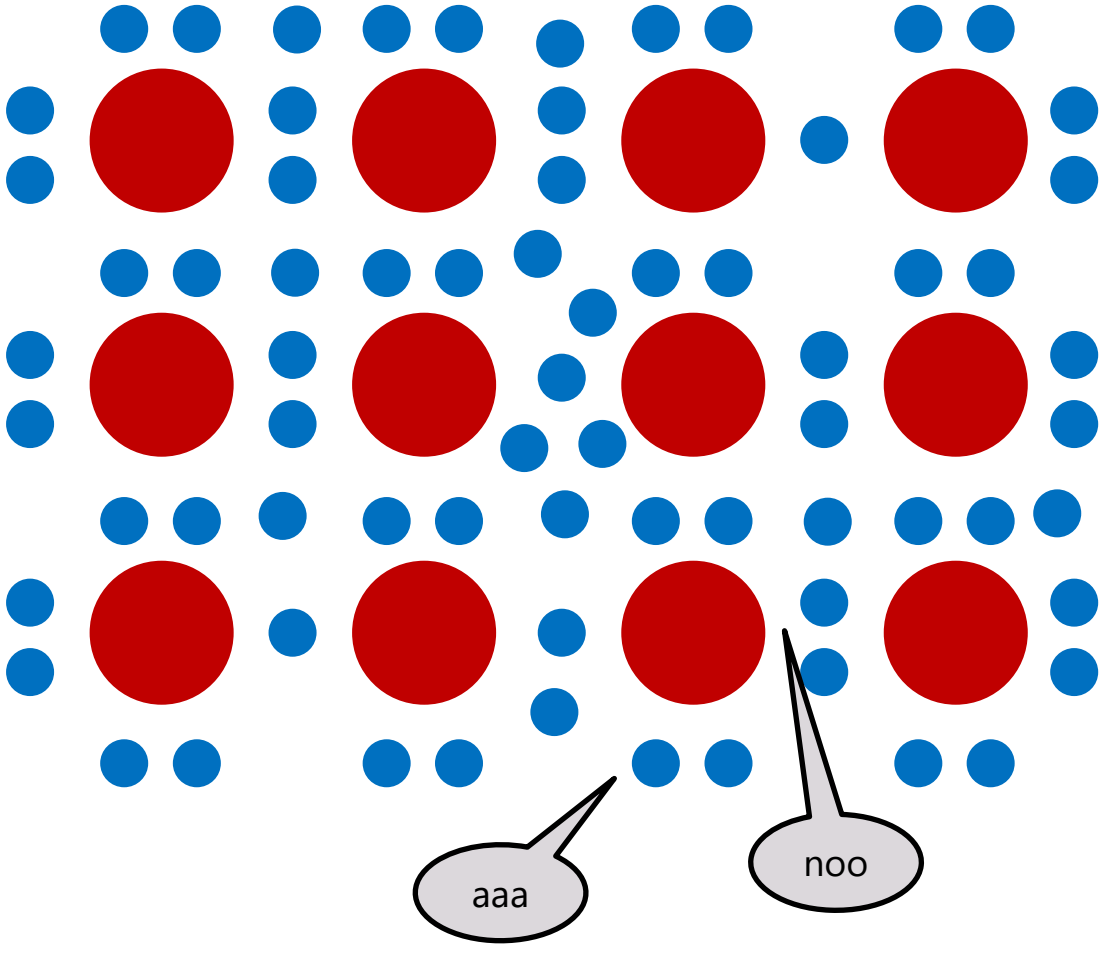


right now, the charges are **balanced:** same number of positive and negative

let's knock it out of whack

Two moles is company, three's a crowd

let's shove more electrons in



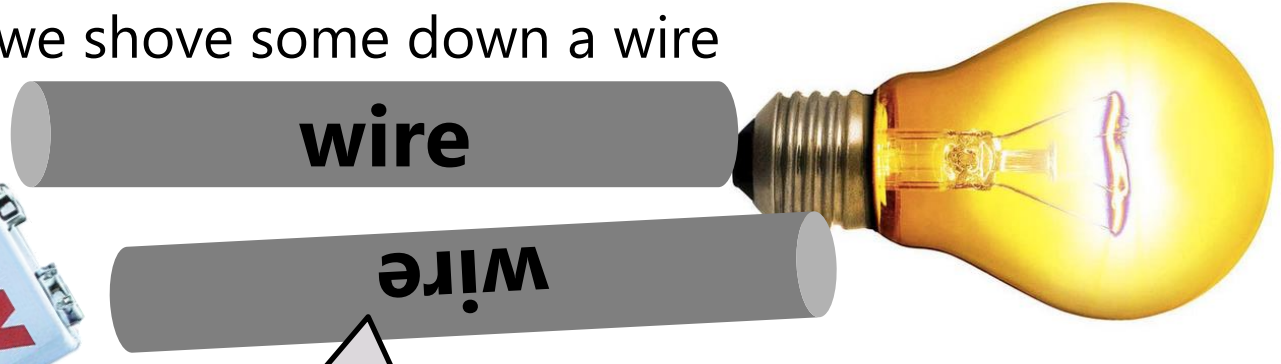
packing more electrons in leads to two things:

1. this metal is now **negatively charged**

2. the electrons are now **closer together** meaning they're **less happy**

Carousel of unhappiness

if we shove some down a wire



and then give them a
(difficult) way out



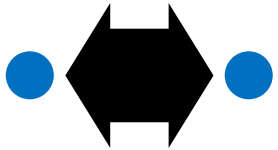
and then have something which will
squeeze them back together again

well now you have electricity

Watch em go

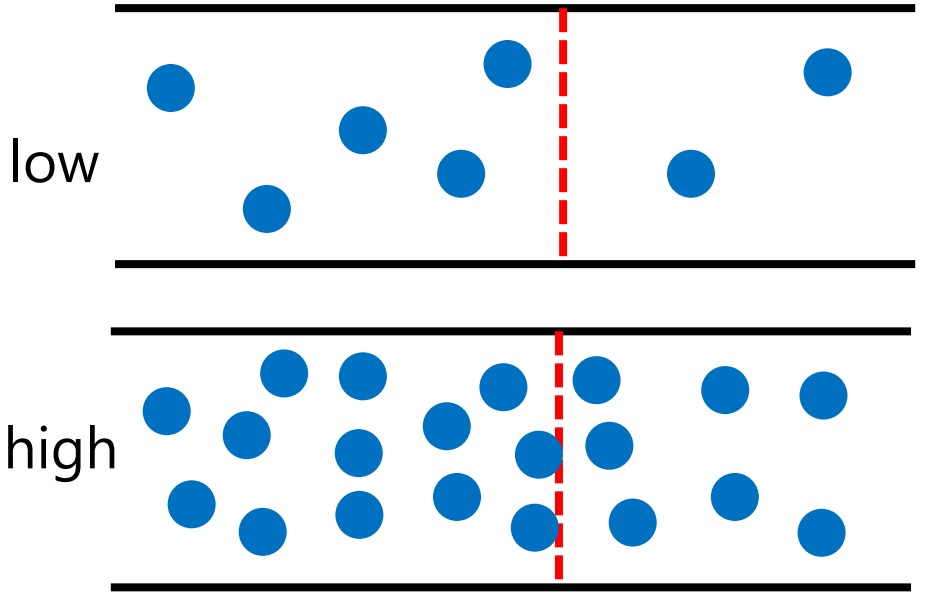
voltage measures electron unhappiness





and that's electricity

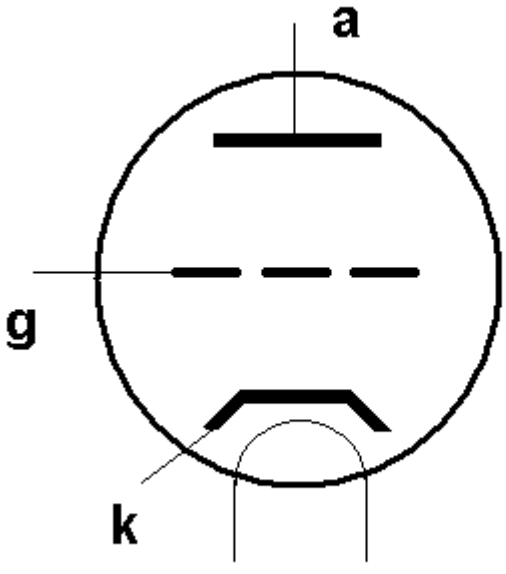
current measures how many electrons per second are moving past a point



Logic Basics

Valves (or tubes) – more fun?

- Why is a vacuum tube called like that?
- They are also called thermionic valves (or simply valves)



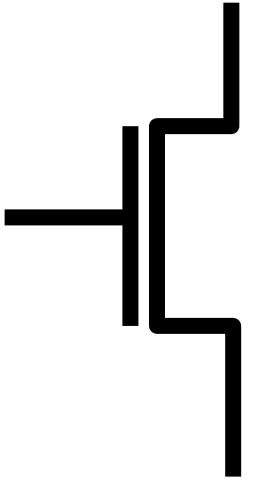


https://en.wikipedia.org/wiki/Vacuum_tube

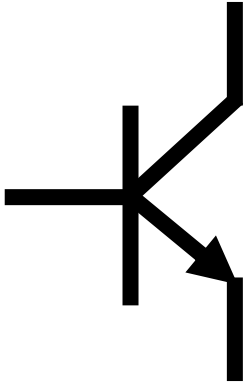
Transistors

- a transistor is like a little valve

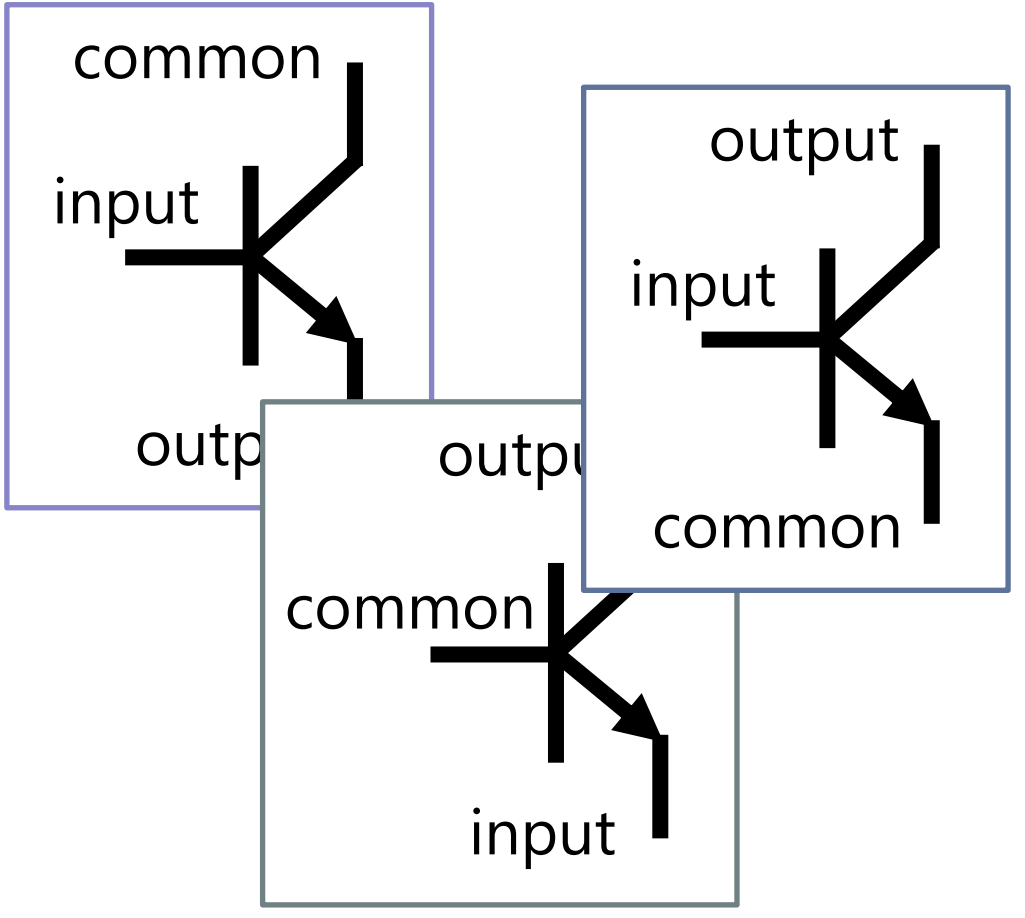
There are multiple types of transistors



MOSFET



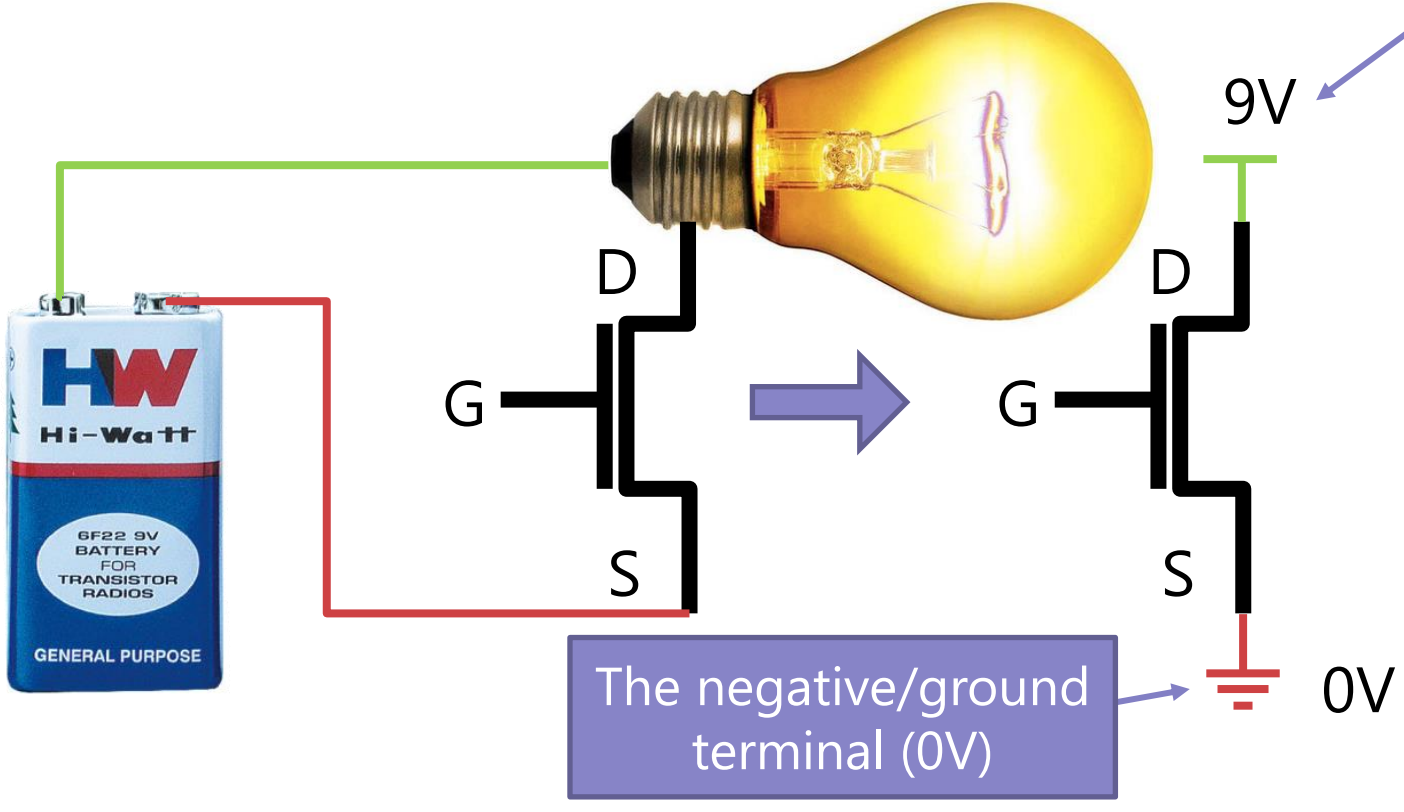
BJT



Can be used many different ways!

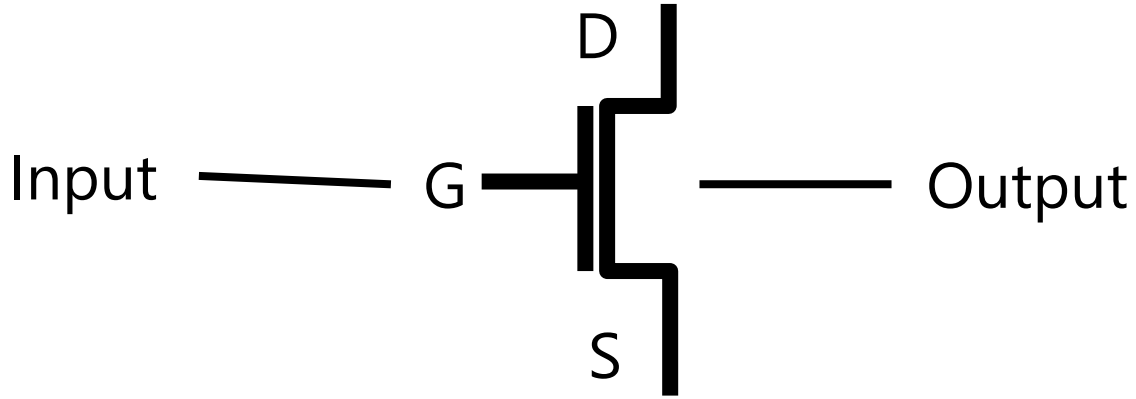
Transistors

- The ones we'll focus on MOSFET transistors
  - Metal-Oxide Semiconductor Field-Effect Transistor
  - Commonly used just as switches (digital electronics)



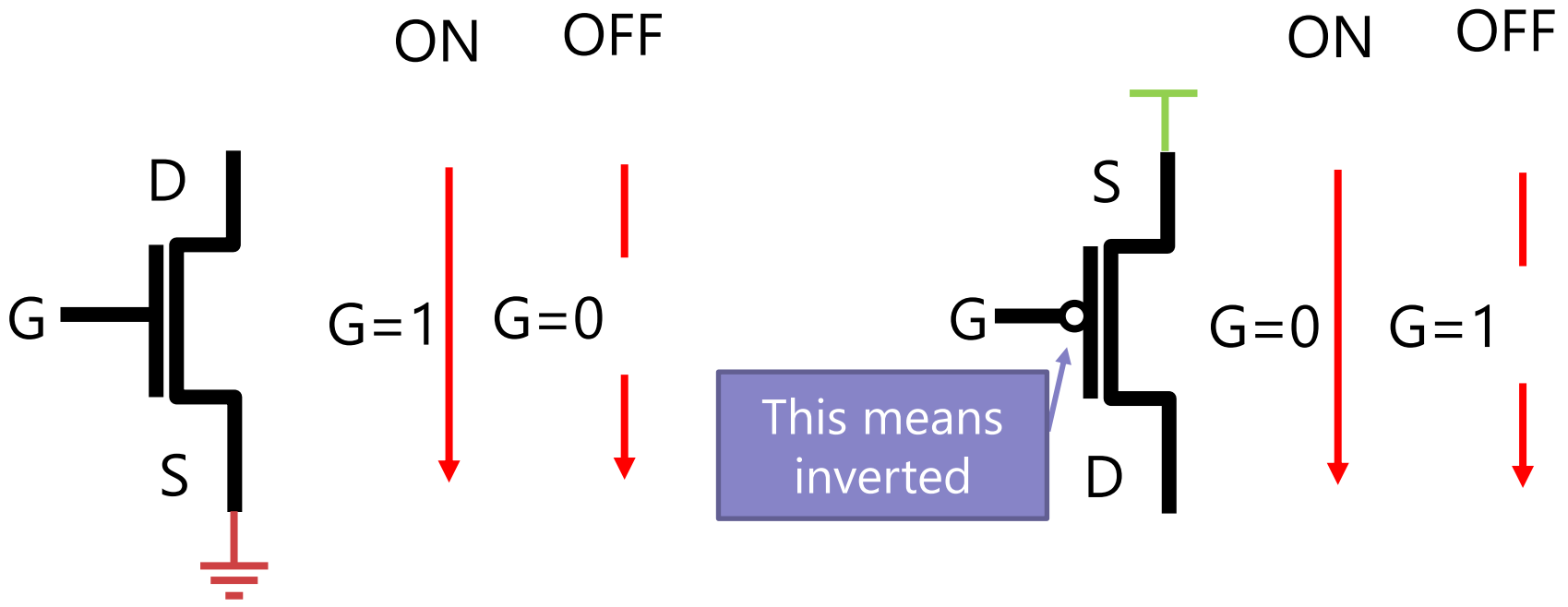
Transistors

- Used as an electrically controlled switch
  - The input is connected to G (Gate)
    - controls the switch state
  - The output is D (drain) is connected to S (source)
- The voltages represent bits! (maybe 9V = **1**, 0V = **0**)



Transistors

- MOSFETs come in two varieties:



"N" Type

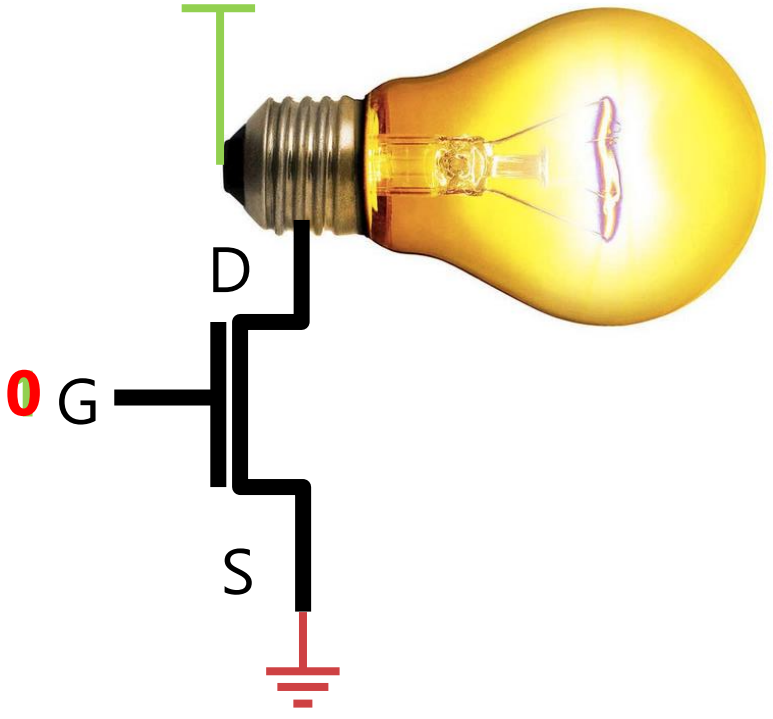
Work when connected to **0**

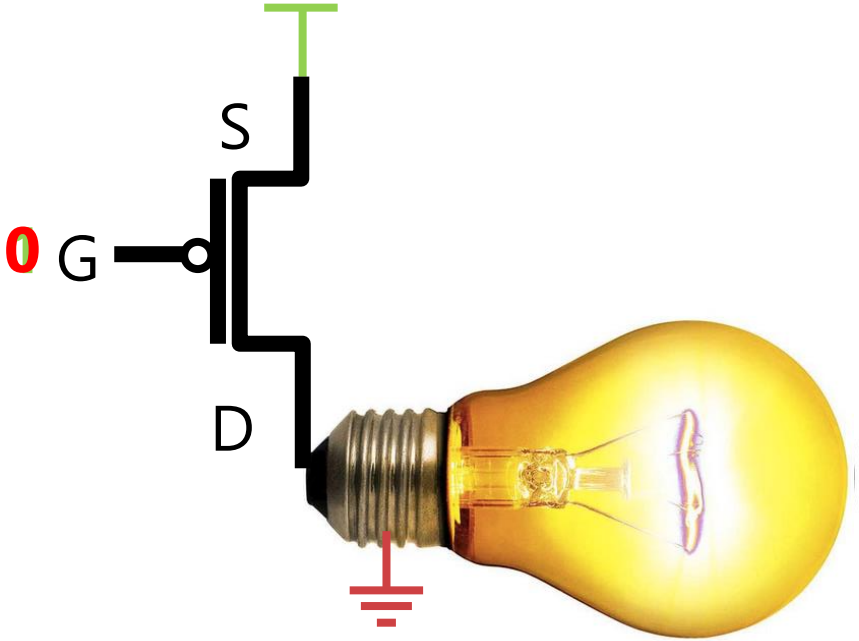
"P" Type

Work when connected to **1**

Transistors – Overly simplified

The current must flow!!



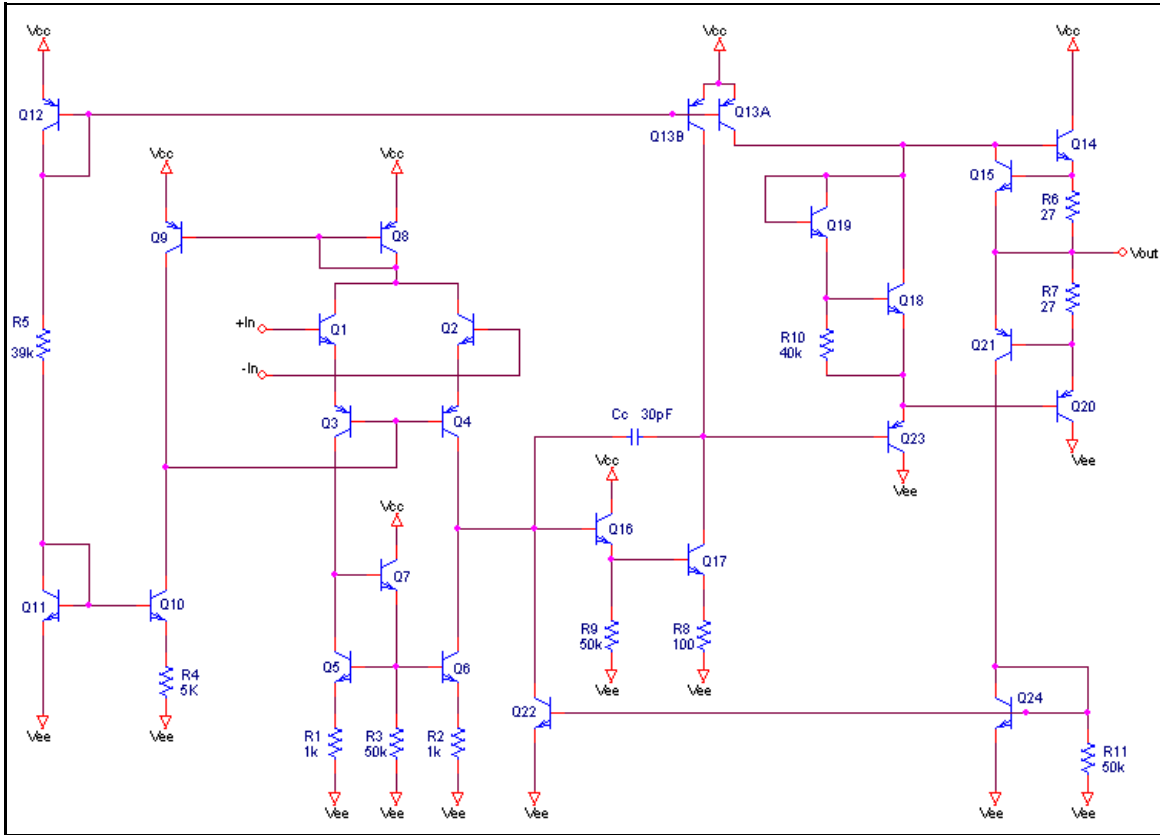


"N" Type

"P" Type

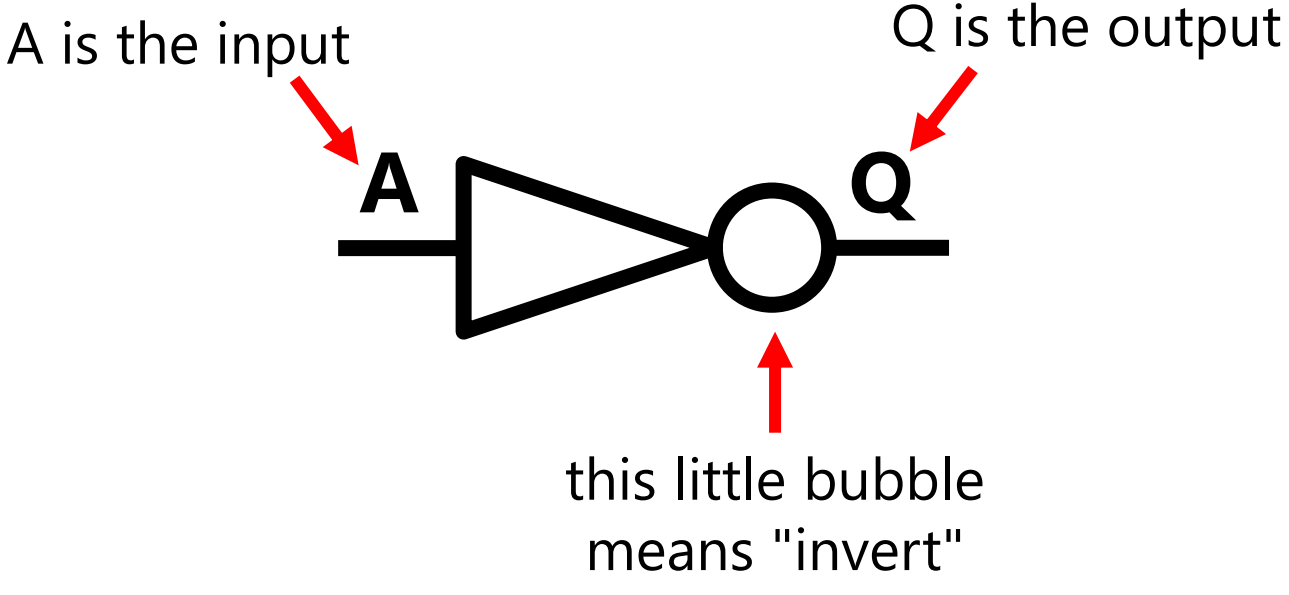
Transistors

Now just put 3 billion of them together! who said EE was hard?



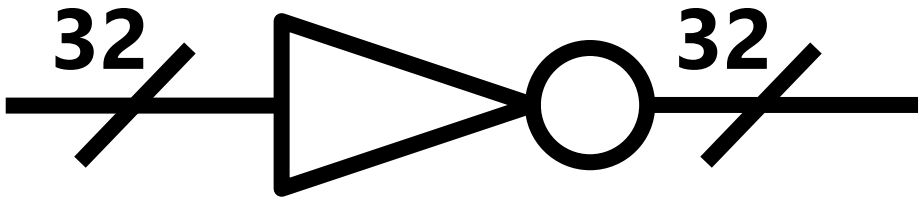
Gates

- We can combine transistors in interesting ways to make **gates**
- A **gate** implements one of the basic **boolean logic** functions
- Let's start with the simplest: the **NOT gate**



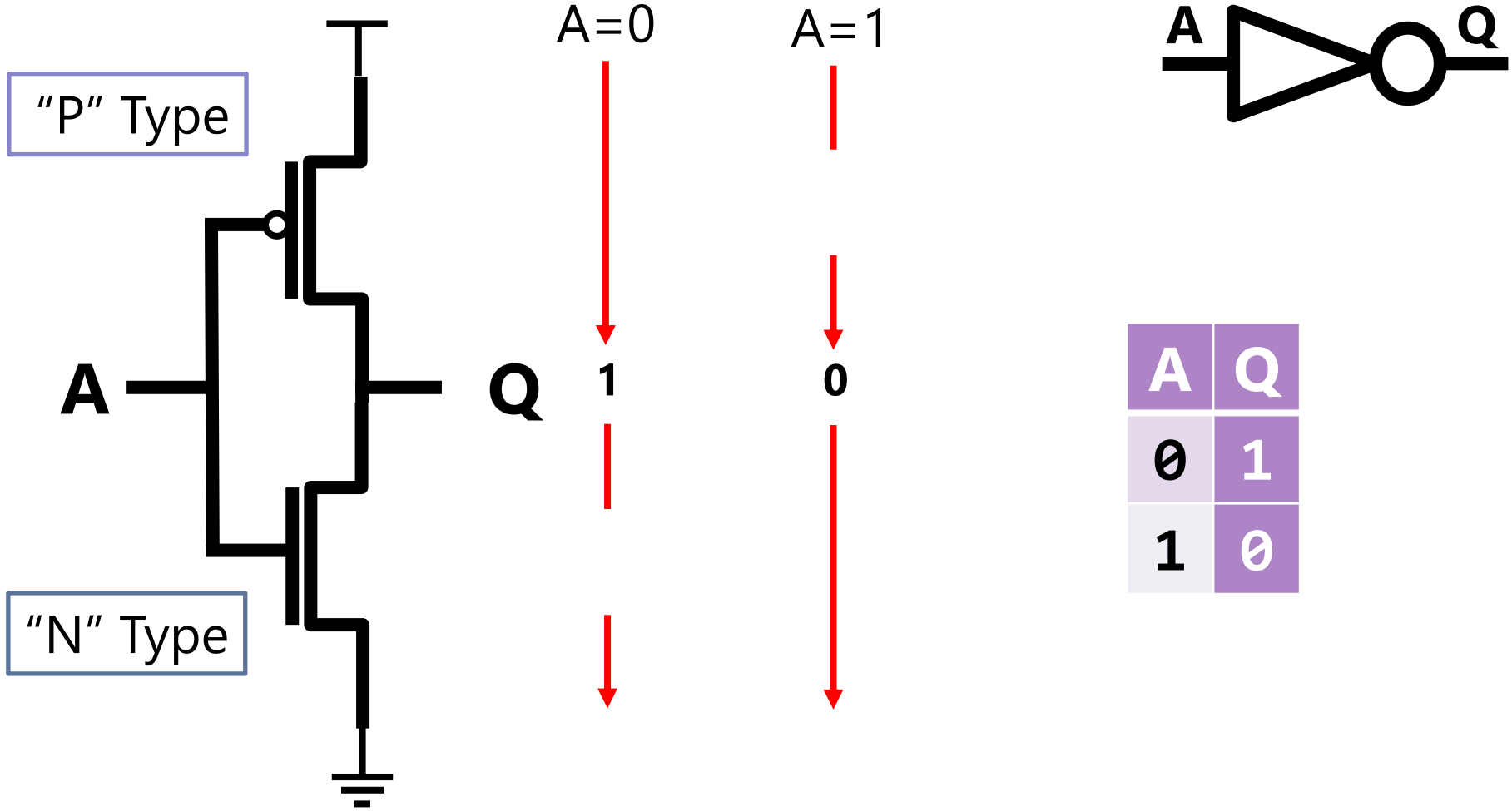
Time to bundle up

- if we want to, say, NOT a 32-bit value, we can draw it like:



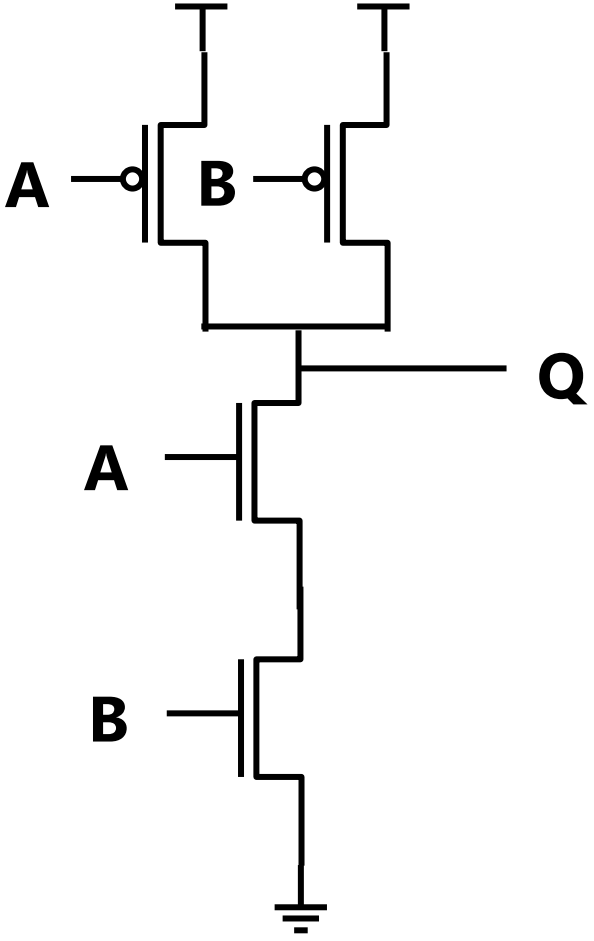
- it's a lot nicer than drawing 32 wires with 32 NOT gates
- Logisim calls these **wire bundles**
  - it doesn't draw the slashes and numbers though…

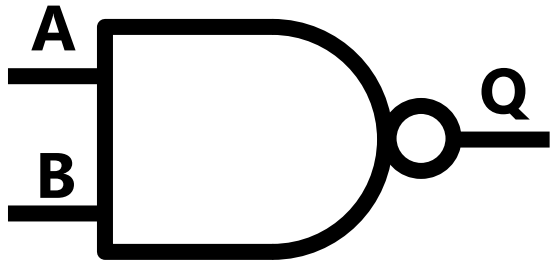
NOT with MOSFETs



A	Q
0	1
1	0

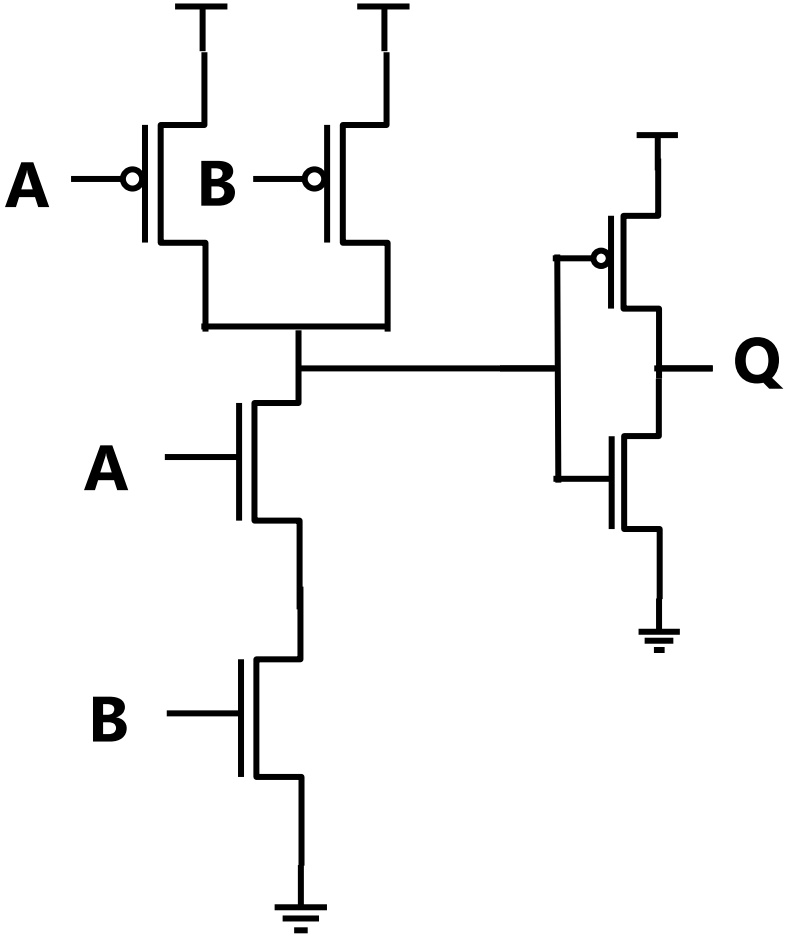
NAND with MOSFETs

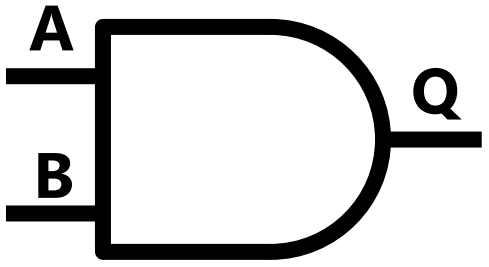




A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

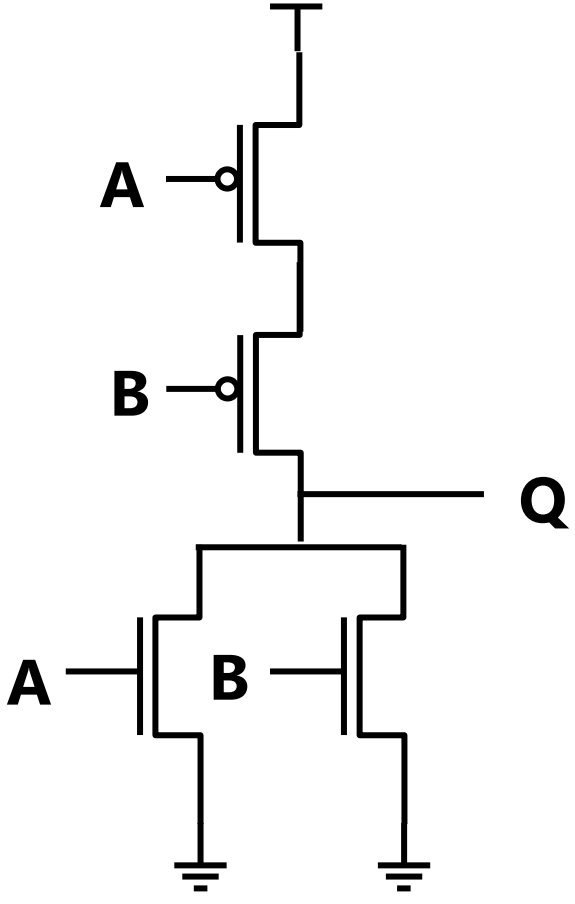
AND with MOSFETs

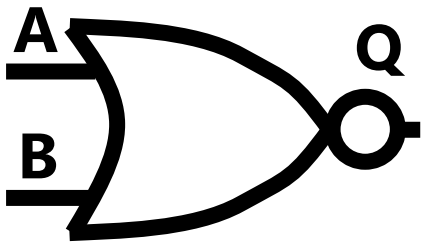




A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

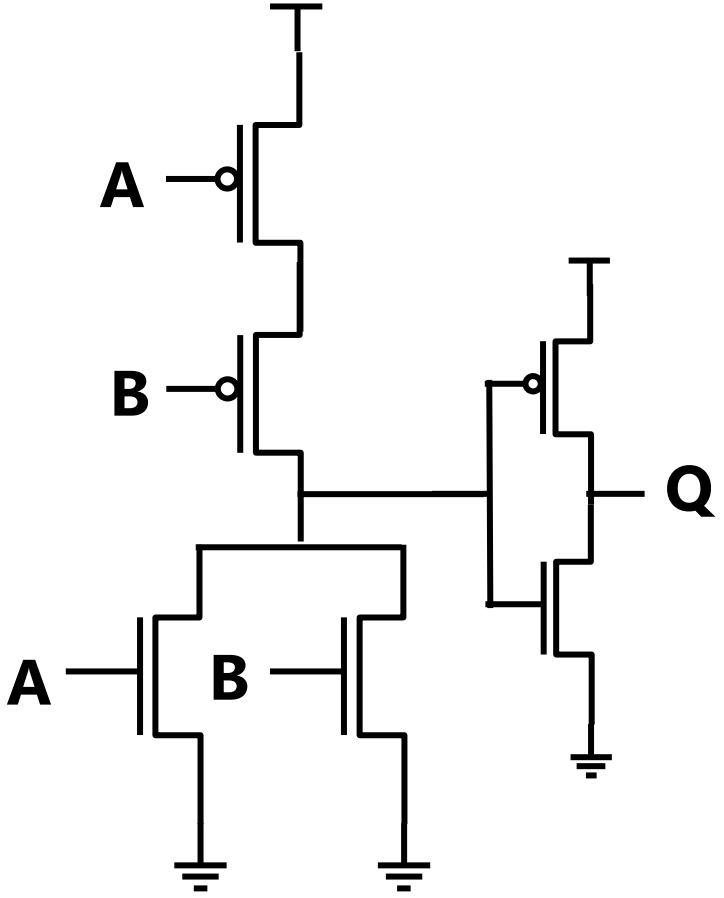
NOR with MOSFETs

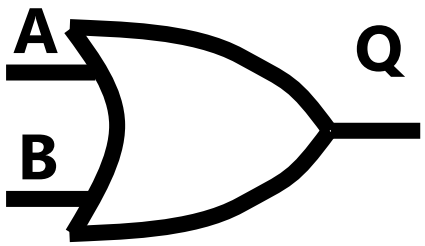




A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

OR with MOSFETs

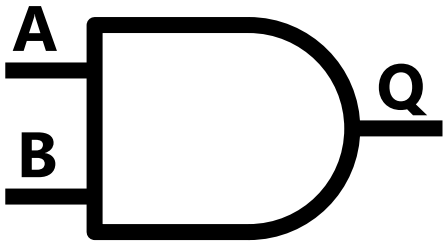




A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

AND, OR, and... XOR?

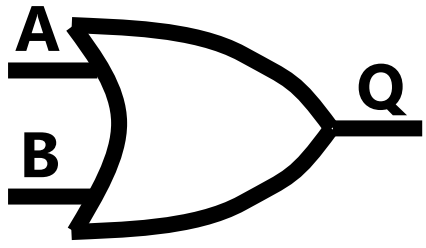
- we know about AND and OR, but what's XOR?



AND gate

A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

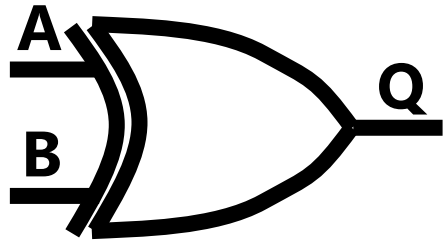
AB



OR gate

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

A+B



XOR gate

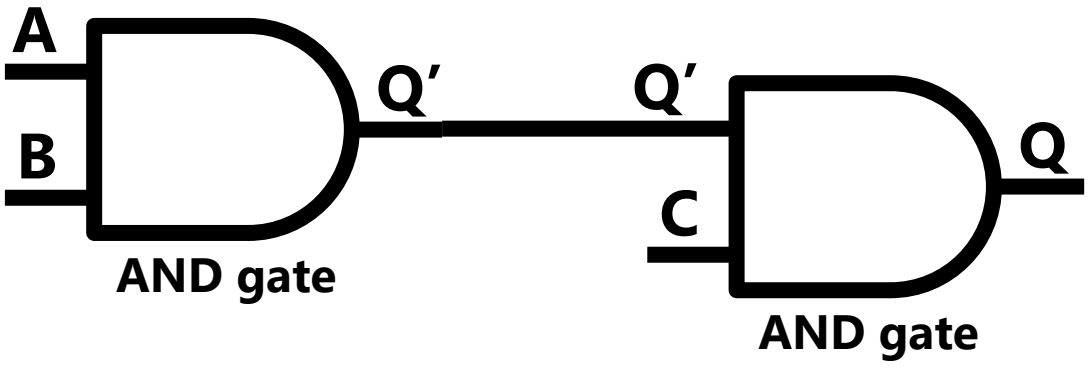
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

A⊕B

e**X**clusive **OR** means "one or the other, but NOT BOTH."

AND (multiple inputs)

- If you have more than 2 inputs…
  you can just concatenate:



A	B	C	Q
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	0
1	1	1	1

AND and/or OR

- Combining logic is just a matter of combining logic gates:

- Look at the truth entries on the truth-table
  - when Q is "1"
- AND each entry variable
- Then, OR each entry:

$$\overline{A}B\overline{C} + \overline{A}BC$$

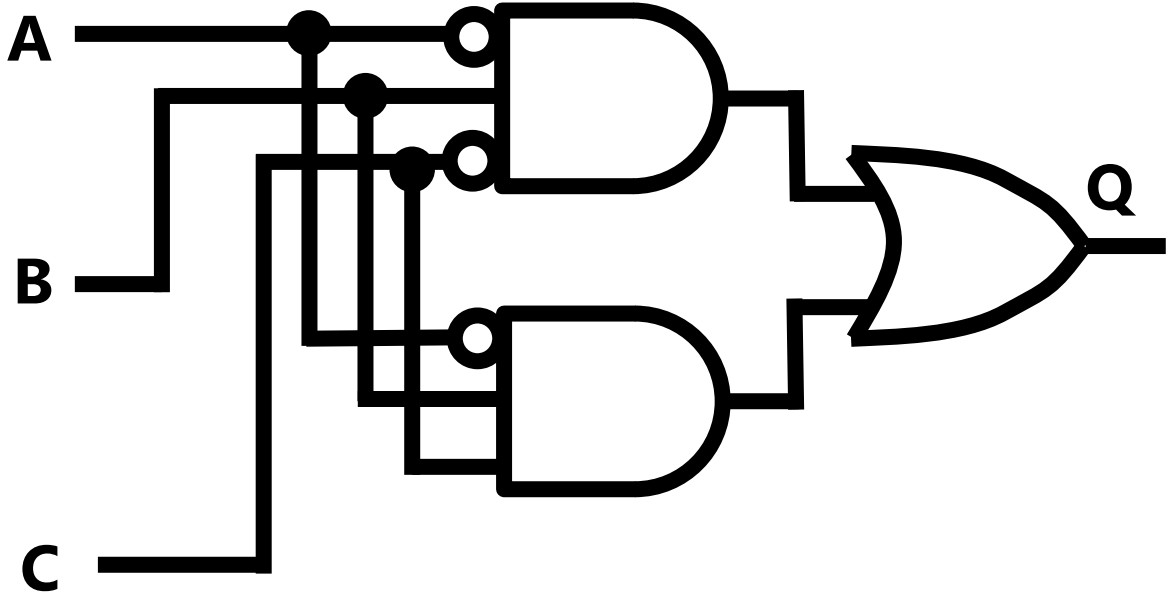
- In English:

It's truth when
"A is false" and "B is true" and "C is false"
OR
"A is false" and "B is true" and "C is true"

A	B	C	Q
0	0	0	0
1	0	0	0
0	1	0	1
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	1
1	1	1	0

AND and/or OR

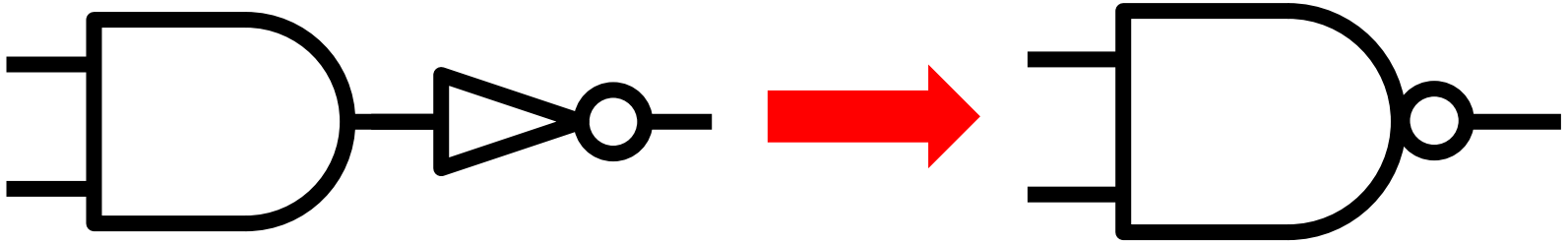
$$\overline{A}B\overline{C} + \overline{A}BC$$



A	B	C	Q
0	0	0	0
1	0	0	0
0	1	0	1
1	1	0	0
0	0	1	0
1	0	1	0
0	1	1	1
1	1	1	0

If you give an electrical engineer a NAND gate...

- if you stick a NOT gate after an AND gate, you get a NAND gate



- this kind of gate has a cool property: it's **universal**
  - in other words, you can build **an entire computer** with NANDs
- but this isn't how real circuits are designed, at least anymore
- digital logic courses use them cause **NAND gate chips are cheap**
- but in Logisim, we have infinite gates for free :D
- **use the kind of gate you need for the situation at hand**